

String Indexing (Exercises)

1. `'python'[0]` 'p'
2. `'python'[1]` _____
3. `'python'[3]` _____
4. `'python'[-1]` _____
5. `'python'[-3]` _____

String Indexing

Strings of text may be surrounded by single quotation marks (').

Putting a number in square brackets ([]) after a string gives the single character at that *index* (position). Indices are zero based, so the first character is at index 0.

Negative indices start from the end of the string, so index -1 is the last character, index -2 is the character before that, and so on.

String Slicing (Exercises)

1. `'abandon'[0:2]` `'ab'`

2. `'abandon'[1:5]` _____

3. `'infinite'[2:100]` _____

4. `'incandescent'[3:8:2]` _____

5. `'airspace'[:3]` _____

6. `'airspace'[3:]` _____

7. `'remote'[4:0:-1]` _____

8. `'duplicate'[:,:]` _____

9. `'straw'[::-1]` _____

String Slicing

Up to three numbers separated by colons (:) within square brackets after a string give a *slice* of the string (a shorter string):

- The first number gives the *start* index of the slice.
- The second number gives the *stop* index. This is exclusive, so the slice includes character up to but not including the stop index.
- The third number gives the *step*. The index is advanced by this much for each character of the slice, so (for example) a step of 2 will include every other character. If the step is negative, the slice works backward from the start to (but not including) the stop.

Any or all of the three numbers specifying a slice may be omitted.

- If no step is provided, the slice starts at the beginning of the string.
 - ... unless the step is negative, in which case the slice starts at the end of the string.
- If no stop is provided, the slice stops at the end of the string.
 - ... unless the step is negative, in which case the slice stops at the beginning of the string.
- If no step is provided, a step of 1 is used. The second colon can be omitted in this case.

If the stop is beyond the end of the string, the slice ends where the string ends.

Arithmetic Operators (Exercises)

1. $2 + 2$ 4
2. $2 + 3$ _____
3. $4 - 1$ _____
4. $3 - 8$ _____
5. $2 * 4$ _____
6. $8 / 2$ _____
7. $7 / 2$ _____
8. $7 // 2$ _____
9. $34 // 10$ _____
10. $34 \% 10$ _____
11. $2 ** 3$ _____
12. $10 ** 20$ _____
13. $2 + 3 * 5$ _____
14. $2 + (3 * 5)$ _____
15. $(2 + 3) * 5$ _____
16. $1 + 2 * 3 - 4 ** 2 / 4$ _____

Arithmetic Operators

Numbers can be combined with familiar arithmetic *operators*. + and – perform addition and subtraction. The multiplication operator is *.

There are three division operators:

- / performs standard division. The answer always includes a decimal point, even if it happens to be an integer.
- // performs integer division, throwing away any remainder.
- %, the *modulo* or *remainder* operator, keeps *only* the remainder.

The ** operator performs exponentiation, so 5 ** 2 computes 5².

Some operators have higher precedence than others, so they are performed first:

- ** has the highest precedence
- *, /, //, and % have the next highest
- + and – have the lowest

Parentheses can be used to ensure that particular operations are performed earlier.

Variables (Exercises)

1. $x = 3$

$x \quad \underline{3}$

2. $x + 2$ _____

3. $y = 4$

y _____

4. $x + y$ _____

5. `color = 'blue'`

`color` _____

6.

```
score = 5  
  
score = 10  
  
score _____
```

7.

a	=	1
b	=	a
a	=	2
b	=	_____

8.
$$\begin{aligned} n &= 10 \\ n &= n + 1 \\ n & \end{aligned}$$

Variables

A *variable* is a name for something. It can be *assigned* a value with the = operator. For example, `x = 3` makes `x` a name for the number 3.

Once it has been assigned a value, the variable can be used in place of the value.

A variable can be redefined by assigning it a new value.

Note that = is not a declaration that two things are the same. It is an assignment that makes the variable name on the left a name for the value on the right.

Assignment makes the variable stand for the *value* on the right side, not the code that produced that value. Specifically:

- If a variable is defined in terms of another variable, later changes to the other variable don't affect the first one.
- A variable can be defined in terms of its current value. For example, `x = x + 1` gets the current value of `x`, adds 1, and makes `x` a name for that new value.

String Operators (Exercises)

1. `'hot' + 'dog'` `'hotdog'`
2. `'shoe' + 'string'` _____
3. `'ma' * 2` _____
4. `'co' * 2 + 'a'` _____
5. `'i' in 'team'` _____
6. `'on' in ('no' * 2)` _____
7. `a = 'brown'`
`b = 'toad'`
`a[1:3] + b[2:]` _____

String Operators

Two strings can be concatenated with `+`.

Multiplying a string by a positive integer (using `*`) concatenates multiple copies of it.

If `a` and `b` are strings, `a in b` is true if the string `a` appears somewhere within `b`.

Quotation Marks (Exercises)

1. "hello" 'hello' _____
2. 'goodbye' _____
3. "don't" _____
4. 'Eddie "The Eagle" Edwards' _____
5. 'You "can\'t" avoid it.' _____
6. "... or \"can\" you?" _____
7. 'Maybe \'sometimes\' you can.' _____

Quotation Marks

Strings may be delimited by either single (') or double (") quotation marks.

Single quotation marks can be used in a string delimited by double quotation marks, and vice versa.

If it is necessary to use the kind of quotation mark that delimits a string inside that string, precede it with a backslash (\).

When it displays strings, Python prefers single quotation marks, but will use double quotation marks to avoid a backslash.

Calling Functions (Exercises)

1. `abs(-3)` 3
2. `abs(12)`
3. `len('python')`
4. `max(2, 3)`
5. `min(2, 3)`
6. `round(3.14159)`
7. `a = 'snake'`
`b = 'language'`
`max(len(a), len(b))`
8. `s = 'python'`
`s[0:len(s)]`
9. `max(1, min(2, 3))`
10. `sorted(s)`
11. `s`

Calling Functions

To *call* a *function*, use the name of the function followed by round parentheses (`()`). If the function takes any *arguments*, put them between the parentheses, separated by commas. The value returned by the function call depends on the function and the arguments.

Here are some of Python's built-in functions:

- `abs(x)` returns the absolute value of the number `x`.
- `len(s)` returns the length of the string `s`.
- `max(x, y)` returns the larger of `x` and `y`.
- `min(x, y)` returns the smaller of `x` and `y`.
- `round(x)` returns the number `x` rounded to the nearest integer. If `x` is exactly halfway between two integers, `round` returns the even one.
- `sorted(s)` returns a sorted version of `s`. It does not modify `s` itself, but returns a new string.

Many of these functions can take additional, optional arguments. Many of them work similarly on other kinds of values, to be explained later.

Lists (Exercises)

1. ['eggs', 'bread', 'tea'] ['eggs', 'bread', 'tea']

```
2. nums = [4, 8, 15, 16, 23, 42]
```

nums[3] _____

3. `nums[2:4]` _____

4. `nums[1] = 99`

nums _____

5. `len(nums)` _____

6. `sorted([4, 1, 3, 2])` _____

```
7. nest = [[6, 7], [1, 1, 3, 8]]
```

nest[1] _____

8. `nest[1][2]` _____

9. left = [1, 2, 3]

```
right = [4, 5, 6]
```

`left + right` _____

10. left * 2 _____

11. 3 in left _____

12. $[2, 3]$ in left _____

Lists

A *list* is a sequence of elements, delimited by square brackets `[` and `]`. The elements might be strings, numbers, or even other lists.

Lists behave much like strings. They can be accessed or sliced just like strings. Functions like `len` and `sorted` work the same way.

Unlike strings, lists can be modified. For example, if `ingredients` is a list, `ingredients[0] = 'olive oil'` replaces the first element of the list.

The `in` operator also behaves slightly differently. For a list `ls`, `a in ls` is true if `a` is an element of `ls`. For a string `s`, `a in s` is true if `a` is a substring of `s`.

Comparisons (Exercises)

1. `1 == 2` False
2. `2 == 2` _____
3. `2 < 3` _____
4. `4 >= 3` _____
5. `4 >= 4` _____
6. `4 >= 5` _____
7. `1 != 2` _____
8. `'a' < 'c'` _____
9. `'a' == 'A'` _____
10. `'a' > 'A'` _____
11. `'abc' < 'xyz'` _____
12. `'rob' < 'robin'` _____
13. `[1, 2, 3] < [1, 3, 5]` _____
14. `[[1, 2], 3] < [[1, 3], 2]` _____
15. `123 < 23` _____
16. `'123' < '23'` _____

Comparisons

There are several operators for comparing things:

- `<` less than
- `<=` less than or equal to
- `==` equal
- `>=` greater than or equal to
- `>` greater than
- `!=` not equal to

The resulting value is either `True` or `False`.

Note the difference between `a = 1`, which means "Make `a` a name for 1.", and `a == 1`, which means "Is `a` equal to 1?".

When comparing sequences like strings and lists, comparisons examine the first element of each. If they are equal, the comparison proceeds to the next element. If a difference is found, or one sequence runs out first, the sequences are not equal.

Characters are compared according to the Unicode specification. Digits are less than upper-case letters, which are in turn less than lower-case letters. Within these groups, smaller digits and earlier letters in the alphabet are considered less.

Types (Exercises)

1.	<code>type(3)</code>	<code><class 'int'></code>
2.	<code>type(3.14159)</code>	
3.	<code>type(True)</code>	
4.	<code>type('hello')</code>	
5.	<code>type([1, 2, 3])</code>	
6.	<code>int(3.14159)</code>	
7.	<code>int('23')</code>	
8.	<code>float(3)</code>	
9.	<code>float('23')</code>	
10.	<code>str(3.14159)</code>	
11.	<code>str([1, 2, 3])</code>	
12.	<code>list('abc')</code>	
13.	<code>bool(0)</code>	
14.	<code>bool(1)</code>	
15.	<code>bool(0.0)</code>	
16.	<code>bool(-1.5)</code>	
17.	<code>bool('false')</code>	
18.	<code>bool('')</code>	
19.	<code>bool([1, 2, 3])</code>	
20.	<code>bool([])</code>	

Types

Every value has a *type*. Conversely, each type is the set of values that have that type.

Types we have seen so far include:

- `int` Integer
- `float` Number including a decimal point or scientific notation, even if it happens to be an integer
- `str` String
- `list` List
- `bool` Boolean logical value: `True` or `False`

The built-in `type` function returns the type of its argument. For example, `type(20)` returns `<class 'int'>`. This is the `int` type; the reason for the more elaborate notation is beyond the scope of this book.

Each type is also the name of a function for converting values of other types. For example, `int('15')`, having been given the string `'15'`, returns the `int` 15.

The type conversion functions generally behave as expected, but:

- Some conversions are not possible. `int('hello')` results in an error.
- Converting a float to an int discards anything after the decimal point.
- `bool` converts any zero or empty value to `False` and any other value to `True`.

Methods (Exercises)

1. a = [3, 1, 4, 2]

```
a.append(5)
```

a [3, 1, 4, 2, 5]

2. `a.extend([6, 7])`

a _____

3. `a.insert(2, 8)`

a _____

4. `a.index(4)`

5. `b = [1, 2, 3, 2, 4, 2, 5]`

b.count(2) _____

6. `b.remove(2)`

b _____

```
7.          b.sort()
```

b _____

8. `'ride a bike'.split()`

() _____

9. `'1,2,3'.split(',')`

```
10. ls = ['eggs', 'bread', 'tea']
```

```
' , '.join(ls)
```

11. `''.join(['a', 'b', 'c'])` _____

Methods

Methods are very similar to functions. Either might return a value or have some side effect like modifying the contents of a list. They are called slightly differently:

- `f(x, y)` is the function `f` being called and given the arguments `x` and `y`. This says to Python, "Hey function `f`, do your thing. You'll need `x` and `y`."
- `x.m(y)` is the method `m` being called *on* the object `x` and given the argument `y`. This says, "Hey object `x`, do `m`. You'll need `y`."

Methods are generally "about" the objects on which they are called.

List methods include:

- `append` modifies the list by adding its argument to the end.
- `extend` modifies the list by adding each element of its argument to the end.
- `insert` inserts an element at a specific index, moving everything after that one position to the right.
- `index` returns the position where the argument first appears in the list.
- `count` returns the number of times the argument appears in the list.
- `remove` removes the first appearance of the argument from the list.
- `sort` modifies the list by sorting it. This is different from the `sorted` function, which creates a new list.

String methods include:

- `split` returns a list of strings found by breaking the string apart. By default, it looks for spaces to break the string into words, but some other separator can be given as an argument.
- `join` creates a string from a list of strings. The string on which the method is called is used as the separator.

List Comprehensions (Exercises)

1. a = [1, 4, 2, 3]
 [i * 2 for i in a] [2, 8, 4, 6]
2. [i + 1 for i in a] _____
3. [5 for i in a] _____
4. [i for i in a if i > 2] _____
5. s = 'python'
 vowels = 'aeiouy'
 [c for c in s if c in vowels] _____

List Comprehensions

A *list comprehension* is a way of converting a list (or other sequence) into another list. The general form is:

```
[expr for var in sequence]
```

The value of this is found by assigning `var` to each element of `sequence` in turn, then evaluating `expr` (an expression which may involve `var`).

Optionally, a list comprehension can also include a condition:

```
[expr for var in sequence if condition]
```

Elements for which this condition is false are omitted from the result.

String Indexing (Solutions)

1. `'python'[0]` 'p'
2. `'python'[1]` 'y'
3. `'python'[3]` 'h'
4. `'python'[-1]` 'n'
5. `'python'[-3]` 'h'

String Slicing (Solutions)

1. `'abandon'[0:2]` `'ab'`
2. `'abandon'[1:5]` `'band'`
3. `'infinite'[2:100]` `'finite'`
4. `'incandescent'[3:8:2]` `'ads'`
5. `'airspace'[:3]` `'air'`
6. `'airspace'[3:]` `'space'`
7. `'remote'[4:0:-1]` `'tome'`
8. `'duplicate'[::]` `'duplicate'`
9. `'straw'[::-1]` `'warts'`

Arithmetic Operators (Solutions)

1. $2 + 2$ 4
2. $2 + 3$ 5
3. $4 - 1$ 3
4. $3 - 8$ -5
5. $2 * 4$ 8
6. $8 / 2$ 4.0
7. $7 / 2$ 3.5
8. $7 // 2$ 3
9. $34 // 10$ 3
10. $34 \% 10$ 4
11. $2 ** 3$ 8
12. $10 ** 20$ 100000000000000000000
13. $2 + 3 * 5$ 17
14. $2 + (3 * 5)$ 17
15. $(2 + 3) * 5$ 25
16. $1 + 2 * 3 - 4 ** 2 / 4$ 3.0

Variables (Solutions)

1. $x = 3$
 x 3
2. $x + 2$ 5
3. $y = 4$
 y 4
4. $x + y$ 7
5. `color = 'blue'`
`color` 'blue'
6. `score = 5`
`score = 10`
`score` 10
7. `a = 1`
`b = a`
`a = 2`
`b` 1
8. `n = 10`
`n = n + 1`
`n` 11

String Operators (Solutions)

1. `'hot' + 'dog'` `'hotdog'`
2. `'shoe' + 'string'` `'shoestring'`
3. `'ma' * 2` `'mama'`
4. `'co' * 2 + 'a'` `'cocoa'`
5. `'i' in 'team'` `False`
6. `'on' in ('no' * 2)` `True`
7. `a = 'brown'`
`b = 'toad'`
`a[1:3] + b[2:]` `'road'`

Quotation Marks (Solutions)

1. "hello" 'hello'
2. 'goodbye' 'goodbye'
3. "don't" "don't"
4. 'Eddie "The Eagle" Edwards' 'Eddie "The Eagle" Edwards'
5. 'You "can\'t" avoid it.' 'You "can\'t" avoid it.'
6. "... or \"can\" you?" '... or "can" you?'
7. 'Maybe \'sometimes\' you can.' "Maybe 'sometimes' you can."

Calling Functions (Solutions)

1. `abs(-3)` 3
2. `abs(12)` 12
3. `len('python')` 6
4. `max(2, 3)` 3
5. `min(2, 3)` 2
6. `round(3.14159)` 3
7. `a = 'snake'`
`b = 'language'`
`max(len(a), len(b))` 8
8. `s = 'python'`
`s[0:len(s)]` 'python'
9. `max(1, min(2, 3))` 2
10. `sorted(s)` ['h', 'n', 'o', 'p', 't', 'y']
11. `s` 'python'

Lists (Solutions)

1. `['eggs', 'bread', 'tea']` `['eggs', 'bread', 'tea']`
2. `nums = [4, 8, 15, 16, 23, 42]`
`nums[3]` `16`
3. `nums[2:4]` `[15, 16]`
4. `nums[1] = 99`
`nums` `[4, 99, 15, 16, 23, 42]`
5. `len(nums)` `6`
6. `sorted([4, 1, 3, 2])` `[1, 2, 3, 4]`
7. `nest = [[6, 7], [1, 1, 3, 8]]`
`nest[1]` `[1, 1, 3, 8]`
8. `nest[1][2]` `3`
9. `left = [1, 2, 3]`
`right = [4, 5, 6]`
`left + right` `[1, 2, 3, 4, 5, 6]`
10. `left * 2` `[1, 2, 3, 1, 2, 3]`
11. `3 in left` `True`
12. `[2, 3] in left` `False`

Comparisons (Solutions)

1.	<code>1 == 2</code>	<u>False</u>
2.	<code>2 == 2</code>	<u>True</u>
3.	<code>2 < 3</code>	<u>True</u>
4.	<code>4 >= 3</code>	<u>True</u>
5.	<code>4 >= 4</code>	<u>True</u>
6.	<code>4 >= 5</code>	<u>False</u>
7.	<code>1 != 2</code>	<u>True</u>
8.	<code>'a' < 'c'</code>	<u>True</u>
9.	<code>'a' == 'A'</code>	<u>False</u>
10.	<code>'a' > 'A'</code>	<u>True</u>
11.	<code>'abc' < 'xyz'</code>	<u>True</u>
12.	<code>'rob' < 'robin'</code>	<u>True</u>
13.	<code>[1, 2, 3] < [1, 3, 5]</code>	<u>True</u>
14.	<code>[[1, 2], 3] < [[1, 3], 2]</code>	<u>True</u>
15.	<code>123 < 23</code>	<u>False</u>
16.	<code>'123' < '23'</code>	<u>True</u>

Types (Solutions)

1.	<code>type(3)</code>	<u><class 'int'></u>
2.	<code>type(3.14159)</code>	<u><class 'float'></u>
3.	<code>type(True)</code>	<u><class 'bool'></u>
4.	<code>type('hello')</code>	<u><class 'str'></u>
5.	<code>type([1, 2, 3])</code>	<u><class 'list'></u>
6.	<code>int(3.14159)</code>	<u>3</u>
7.	<code>int('23')</code>	<u>23</u>
8.	<code>float(3)</code>	<u>3.0</u>
9.	<code>float('23')</code>	<u>23.0</u>
10.	<code>str(3.14159)</code>	<u>'3.14159'</u>
11.	<code>str([1, 2, 3])</code>	<u>'[1, 2, 3]'</u>
12.	<code>list('abc')</code>	<u>['a', 'b', 'c']</u>
13.	<code>bool(0)</code>	<u>False</u>
14.	<code>bool(1)</code>	<u>True</u>
15.	<code>bool(0.0)</code>	<u>False</u>
16.	<code>bool(-1.5)</code>	<u>True</u>
17.	<code>bool('false')</code>	<u>True</u>
18.	<code>bool('')</code>	<u>False</u>
19.	<code>bool([1, 2, 3])</code>	<u>True</u>
20.	<code>bool([])</code>	<u>False</u>

Methods (Solutions)

1. `a = [3, 1, 4, 2]`
 `a.append(5)`
 `a` `[3, 1, 4, 2, 5]`
2. `a.extend([6, 7])`
 `a` `[3, 1, 4, 2, 5, 6, 7]`
3. `a.insert(2, 8)`
 `a` `[3, 1, 8, 4, 2, 5, 6, 7]`
4. `a.index(4)` `3`
5. `b = [1, 2, 3, 2, 4, 2, 5]`
 `b.count(2)` `3`
6. `b.remove(2)`
 `b` `[1, 3, 2, 4, 2, 5]`
7. `b.sort()`
 `b` `[1, 2, 2, 3, 4, 5]`
8. `'ride a bike'.split()` `['ride', 'a', 'bike']`
9. `'1,2,3'.split(',')` `['1', '2', '3']`
10. `ls = ['eggs', 'bread', 'tea']`
 `', '.join(ls)` `'eggs, bread, tea'`
11. `''.join(['a', 'b', 'c'])` `'abc'`

List Comprehensions (Solutions)

1. a = [1, 4, 2, 3]
 [i * 2 for i in a] [2, 8, 4, 6]
2. [i + 1 for i in a] [2, 5, 3, 4]
3. [5 for i in a] [5, 5, 5, 5]
4. [i for i in a if i > 2] [4, 3]
5. s = 'python'
 vowels = 'aeiouy'
 [c for c in s if c in vowels] ['y', 'o']